

Supplementary Materials for
**All-solution-processed scalable and wearable organic thermoelectrics by
structurally mimicking transverse thermoelectric effects**

Juhyung Park *et al.*

Corresponding author: Jeonghun Kwak, jkwak@snu.ac.kr

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This PDF file includes:

Supplementary Notes S1 to S5
Figs. S1 to S19
Tables S1 and S2
References

Note S1. Thermal-circuit and geometry design framework for dual- κ substrates

In a dual- κ substrate, a high-thermal-conductivity region (κ_{high}) and a low-thermal-conductivity (κ_{low}) region are placed adjacent to each other laterally. When the substrate is heated from below under a near-isothermal boundary, heat predominantly flows vertically within each region, resulting in different surface temperatures: a higher surface temperature $T_{\text{s,h}}$ on the high- κ side and a lower surface temperature $T_{\text{s,c}}$ on the low- κ side. This vertical heat-conduction asymmetry establishes a lateral surface temperature difference $\Delta T_{\parallel} = T_{\text{s,h}} - T_{\text{s,c}}$. In the absence of any lateral heat exchange, this vertical effect alone would define the surface temperature difference.

However, in practice, lateral thermal coupling between the two regions partially relaxes this temperature difference. To isolate this lateral redistribution mechanism, we introduce a thermal-resistor model that considers only in-plane heat conduction while treating the vertical temperature contrast as a fixed boundary condition. The lateral heat-flow path consists of serial conduction through the two regions, along with a finite thermal contact resistance $R_{\text{tc}} = 1/(h_{\text{c}}A)$ (K W^{-1}) (53), where h_{c} is the thermal contact conductance and A is the cross-sectional area. The total lateral thermal resistance is therefore, $R_{\text{lat}} = R_{\text{high}} + R_{\text{tc}} + R_{\text{low}}$, where $R_{\text{high(low)}} = L_{\text{high(low)}}/(\kappa_{\text{high(low)}}A)$, where $L_{\text{high(low)}}$ represent the effective lateral conduction lengths in the κ_{high} and κ_{low} regions, respectively. Assuming identical cross-sections and uniform top-surface convection, the retained lateral temperature contrast can be expressed relative to the contrast that would exist in the absence of any lateral heat exchange. We denote this idealized vertical-transport-limited contrast as $\Delta T_{\parallel}^{\text{intrinsic}}$, corresponding to the case $R_{\text{tc}} \rightarrow \infty$. Under these assumptions, the retained temperature contrast scales approximately as

$$\frac{\Delta T_{\parallel}(R_{\text{tc}})}{\Delta T_{\parallel}^{\text{intrinsic}}} \approx \frac{R_{\text{tc}}}{R_{\text{high}} + R_{\text{tc}} + R_{\text{low}}} \quad (\text{S1})$$

Therefore, increasing R_{tc} suppresses lateral heat exchange and preserves a larger fraction of the intrinsic κ -driven surface temperature difference originally generated by vertical thermal transport. In the limit $R_{\text{tc}} \rightarrow \infty$, the two regions become laterally thermally isolated and the intrinsic contrast is fully retained, whereas $R_{\text{tc}} = 0$ corresponds to perfect lateral thermal coupling and complete relaxation of $\Delta T_{\parallel}$.

While the scaling relation above highlights the role of R_{tc} in preserving the κ -driven surface contrast, the spatial extent over which this enhancement is expressed depends on the device geometry. In particular, the interplay between the lateral conducting length L_{D} and the thermal healing length ℓ governs how $\Delta T_{\parallel}$ manifests spatially. The thermal healing length ℓ represents the distance over which a lateral surface-temperature gradient decays due to the competition between in-plane thermal conduction and heat loss to the environment (57). For a thin substrate, this length scale can be estimated as

$$\ell \sim \sqrt{\frac{\kappa_{\text{eff}} t_{\text{sub}}}{h_{\text{eff}}}} \quad (\text{S2})$$

where t_{sub} is the substrate thickness, κ_{eff} is the effective in-plane thermal conductivity of the dual- κ medium, and h_{eff} denotes the effective convective/radiative heat transfer coefficient to the surroundings (58). When the lateral conducting length is comparable to or smaller than the thermal healing length ($L_{\text{D}} \leq \ell$), the interface region dictates the surface temperature distribution, and increasing R_{tc} amplifies the observed $\Delta T_{\parallel}$ across the entire device. In contrast, when the lateral conducting length greatly exceeds the healing length ($L_{\text{D}} \gg \ell$), the global temperature drop remains primarily governed by the κ -contrast, while R_{tc} only sharpens and extends the local

interfacial gradient without affecting the far-field temperature difference dictated by the vertical κ -contrast. Thus, R_{tc} does not create the thermal contrast itself; rather, it controls how effectively the intrinsic κ -induced contrast is retained over a finite lateral distance. In practical device design, these results imply that the lateral geometry and thermal contact resistance should be co-optimized. In particular, selecting an electrode spacing L_E on the order of the thermal healing length ℓ maximizes the expression of the κ -induced temperature difference, while engineering sufficient thermal contact resistance R_{tc} suppresses in-plane thermal relaxation. Within fabrication and packaging constraints, targeting $L_E \sim \ell$ and minimizing unintended thermal bridging at the interface provide a general design guideline for maximizing the pseudo-transverse thermoelectric response in dual- κ substrates.

To verify the physical trends predicted by this model, we also performed steady-state FEA simulations (fig. S3). The numerical results are fully consistent with the analytical picture: (i) increasing thermal contact resistance preserves κ -induced contrast over longer distances, and (ii) increasing κ contrast increases the intrinsic vertical temperature asymmetry. Details of the simulation framework and boundary conditions are presented in Note S4.

Note S2. Interfacial fracture energy of the Cu–PDMS/PDMS joint

We quantified the interfacial fracture energy (Γ) between Cu–PDMS and PDMS using a Rivlin–Thomas–type protocol with paired unnotched/notched tensile samples as shown in fig. S6 (59, 60). Experiments were performed on an Instron 3367 (500 N load cell; peel rate 50 mm min⁻¹). For each pair, the unnotched sample was pulled in tension to obtain the force-clamp distance curve $F(L)$. The notched sample (interfacial notch at the junction) was tested under the same setup; the critical clamp distance L_c at which the notch transitions to a running interfacial crack (onset of the sharp post-peak force drops). The mechanical work done to the unnotched specimen up to L_c is

$$U(L_c) = \int_0^{L_c} F(L) dL \quad (S3)$$

The interfacial fracture energy is then obtained by normalizing the work with the initial width a_0 and thickness b_0 of the notched specimen:

$$\Gamma = \frac{U(L_c)}{a_0 b_0} \quad (S4)$$

For our samples, the initial width and thickness were $a_0 = 16.3$ mm and $b_0 = 0.49$ mm, respectively. The critical clamp distance was $L_c = 6.594$ mm, and integrating the unnotched F – L curve up to L_c gave $U(L_c) = 21.55$ N mm. Using equation (S4), this yields an $\Gamma \approx 2700$ J m⁻². This value is substantially higher than typical adhesion energies reported for untreated polymer interfaces (10–100 J m⁻²) (54) or even chemically treated polymer/metal junctions (100–500 J m⁻²). (61) Values exceeding 1000 J m⁻² are generally regarded as indicative of strong interfacial bonding. (62) Thus, the Cu–PDMS and PDMS joint exhibits robust adhesion that resists crack propagation under routine handling and bending.

Note S3. Transient analysis of surface temperature response

The heating and relaxation dynamics of the Cu–PDMS and PDMS regions were analyzed using a step-wise bi-exponential model:

$$T(t) = \begin{cases} T_0, & t < t_{\text{on}} \\ T_0 + (T_{\text{ss}} - T_0) \left[1 - \left(ae^{-\frac{t}{\tau_{1h}}} + (1-a)e^{-\frac{t}{\tau_{2h}}} \right) \right], & t_{\text{on}} \leq t < t_{\text{off}} \\ T_0 + (T(t_{\text{off}}) - T_0) \left[ae^{-\frac{t-t_{\text{off}}}{\tau_{1r}}} + (1-a)e^{-\frac{t-t_{\text{off}}}{\tau_{2r}}} \right], & t \geq t_{\text{off}} \end{cases} \quad (\text{S5})$$

where T_0 denotes the initial temperature, T_{ss} the steady-state surface temperature during heating, t_{on} , and t_{off} the switching times for heating-on and heating-off, a the weighting factor between fast (τ_1) and slow (τ_2) processes, and τ_{1h} , τ_{2h} , τ_{1r} , τ_{2r} the characteristic time constants for heating and relaxation, respectively. As shown in fig. S8, both Cu–PDMS and pristine PDMS exhibit a rapid rise upon heating and a subsequent thermal relaxation once the Peltier is switched off, which are well described by the above model. Importantly, the surface temperatures remain steady up to the switching time t_{off} under a constant heat source at $T_H = 66^\circ\text{C}$, with no evidence of thermal drift. The temperature difference stabilized at $\Delta T_{\parallel} \approx 8.2^\circ\text{C}$ determined from the last 10% of the raw data points. The extracted parameters (Table S1) indicate two distinct thermal timescales: a fast response (τ_1) associated with interfacial heating and a slower component (τ_2) reflecting bulk thermal diffusion. Compared to pristine PDMS, Cu–PDMS shows a higher steady-state temperature and stabilizes faster, highlighting the beneficial role of conductive filler. Longer-duration monitoring at a lower heat-source setting ($T_H = 60^\circ\text{C}$, up to ~ 2500 s) likewise demonstrates a stable $\Delta T_{\parallel}$ without drift; see fig. S9. Overall, this analysis demonstrates that the dual- κ substrate not only establishes a temperature difference quickly but also maintains it robustly over extended heating periods.

Note S4. FEA simulation of the pT-TEG devices

A steady-state FEA simulation was performed to evaluate the heat transfer characteristics of the pT-TEG devices and to compare the resulting temperature distribution with experimental results. A multilayer 3D geometry was constructed, consisting of the Peltier module, the dual- κ substrate composed of Cu-PDMS ($\kappa_{\text{high}} = 0.88 \text{ W m}^{-1} \text{ K}^{-1}$) and pristine PDMS ($\kappa_{\text{low}} = 0.15 \text{ W m}^{-1} \text{ K}^{-1}$), an SWCNT thermoelectric layer, and flexible Ag-composite electrodes. All geometrical parameters and materials properties (including thermal conductivity, electrical conductivity) were set to match the fabricated devices. In the experiment, heating is supplied by an external Peltier module positioned beneath the substrate. To emulate this configuration, a solid Ag block was placed directly below the substrate in the model. The bottom surface of this Ag block was fixed to the experimentally measured heater temperature for each device configuration (i.e., $T_{\text{H}} = 60$ or $66 \text{ }^{\circ}\text{C}$), while no other surface was held at a fixed temperature. Convective heat transfer ($h = 15 \text{ W m}^{-2} \text{ K}^{-1}$) and surface-to-ambient radiation were applied to the top and lateral surfaces, using material-specific emissivity for the electrodes, SWCNT, Cu-PDMS, and PDMS regions. Interfacial heat transport across the κ -contrast boundary was modeled using an aerial thermal contact resistance of $R'_{\text{tc}} = 1 \times 10^{-4} \text{ m}^2 \text{ K W}^{-1}$, representative of polymer-polymer and polymer-composite interfaces. This value falls within the literature-reported range (10^{-5} – $10^{-3} \text{ m}^2 \text{ K W}^{-1}$) (63–65) and its influence on $\Delta T_{\parallel}$ is evaluated in fig. S3. A representative steady-state temperature distribution for a single-cell pT-TEG at $T_{\text{H}} = 66 \text{ }^{\circ}\text{C}$ is shown in fig. S10. The simulations for 8 pairs pT-TEG modules presented in the main text (Fig. 5B) were conducted under the same modeling assumptions and boundary conditions with the heat-source temperature set to $T_{\text{H}} = 60 \text{ }^{\circ}\text{C}$ to match the experimental measurement. For the 8-pair module, the Electrical Currents interface was also enabled, grounding one electrode and leaving the other floating to obtain the open-circuit voltage.

Note S5. Use of IR thermography and thermocouple measurements for temperature analysis

An infrared camera was employed in this study to visualize the spatial temperature distribution and the temporal stability of the in-plane temperature gradient during device operation. The IR images shown in Fig. 4E provide a qualitative view of the temperature distribution across the Cu–PDMS and PDMS regions and confirm the formation of a stable in-plane temperature gradient near the κ -contrast interface. However, IR camera–based temperature measurements are influenced by experimental factors such as the finite spatial resolution of the detector relative to the device feature size and emissivity differences between constituent materials. In the present pT-TEG devices, the emissivity contrast between the Cu–PDMS and pristine PDMS regions, together with spatial averaging inherent to IR imaging, can lead to smoothing of the apparent temperature gradient near the interface. For this reason, thermocouple measurements were used as the primary source of quantitative temperature values for device performance analysis and for validation of FEA simulations. As shown in Fig. S10, the in-plane temperature profiles measured by thermocouples show good agreement with the FEA-predicted temperature distribution under identical heating conditions, confirming that the thermal model accurately captures the experimentally observed temperature difference.

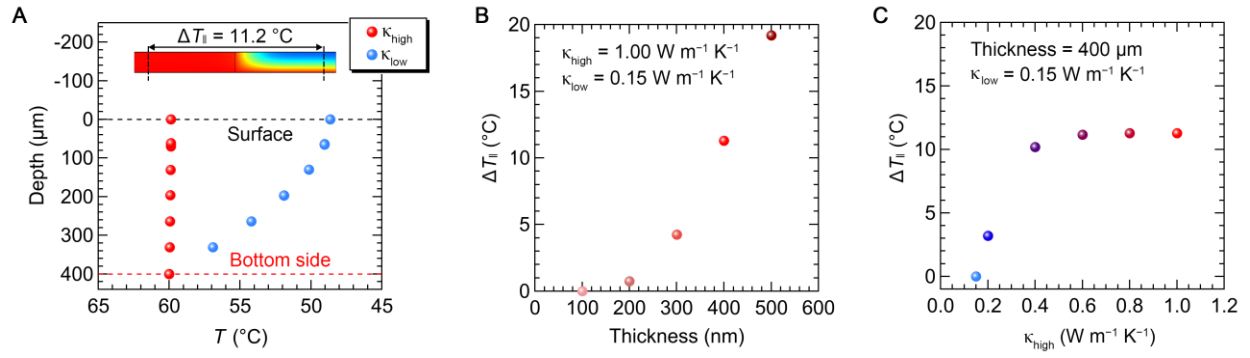


Fig. S1.

Finite element analysis (FEA) simulations of heat transfer on the dual-κ substrate. (A)

Depth profile of the of the dual-κ substrates, showing a substantial temperature decrease from the bottom to the surface in the κ_{low} region, while the temperature remains nearly constant in the κ_{high} region. The resulting in-plane temperature gradient (ΔT_{||}) is 11.2 °C. **(B)** The ΔT_{||} of the dual-κ substrate plotted as a function of the κ_{high}, and **(C)** as a function of substrate thickness. In both cases, κ_{low} is fixed at 0.15 W m⁻¹ K⁻¹.

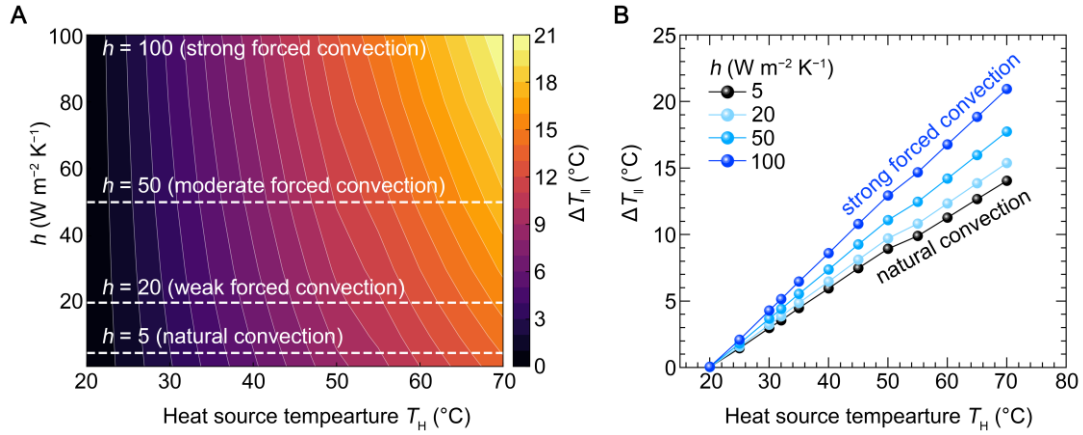


Fig. S2.

Effects of heat transfer coefficient h and heat source temperature T_H on in-plane temperature gradient $\Delta T_{||}$. (A) Contour plot of $\Delta T_{||}$ as a function of the h and the heat source temperature (T_H). The dashed white lines indicate representative cases: $h = 5 \text{ W m}^{-2} \text{K}^{-1}$ (natural convection), $h = 20 \text{ W m}^{-2} \text{K}^{-1}$ (weak forced convection), $h = 50 \text{ W m}^{-2} \text{K}^{-1}$ (moderate forced convection), and $h = 100 \text{ W m}^{-2} \text{K}^{-1}$ (strong forced convection). (B) Line-cut profiles of $\Delta T_{||}$ versus T_H at these h values, showing that $\Delta T_{||}$ monotonically increases with both h and T_H .

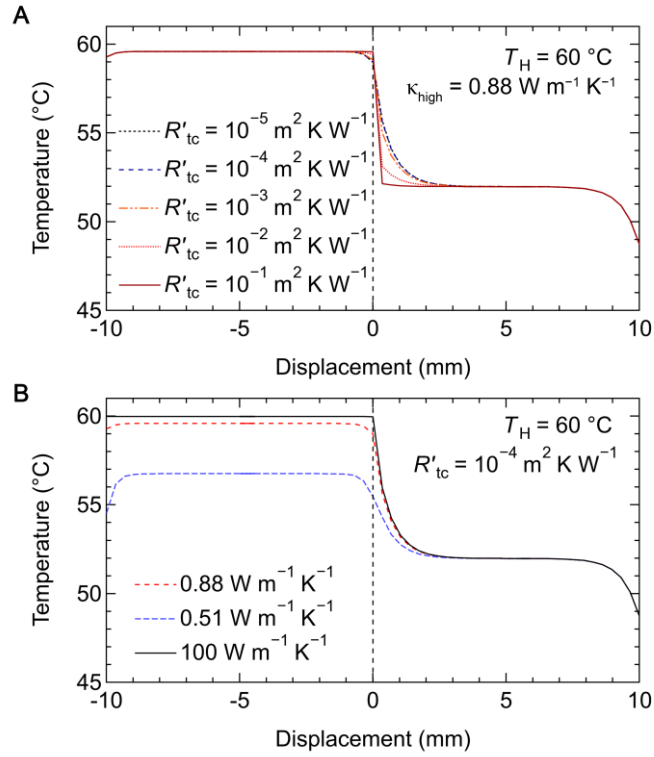


Fig. S3.

FEA simulations validating the dual- κ model. (A) Increasing areal thermal contact resistance R'_{tc} suppresses lateral heat spreading and preserves κ -induced contrast. (B) Increasing κ_{high} increases the intrinsic vertical temperature asymmetry. In both cases, far-field temperatures converge, consistent with the analytical prediction that vertical κ contrast generates the temperature gradient, while R'_{tc} governs its lateral retention. R'_{tc} is the area-normalized thermal contact resistance, defined as $R'_{tc} = R_{tc}/A$.

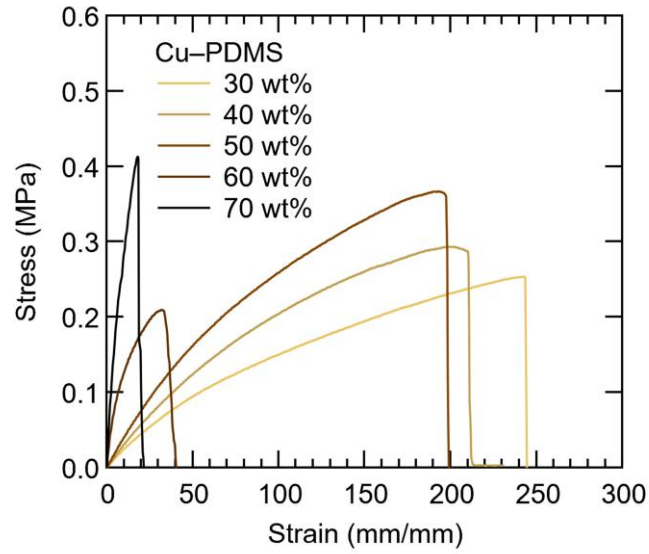


Fig. S4.

Mechanical properties of Cu-PDMS composites. Stress-strain curves for Cu-PDMS composites with different weight ratios of Cu microparticles are presented. The modulus increases as a function of the Cu microparticle weight ratio, while still maintaining elastic deformation.

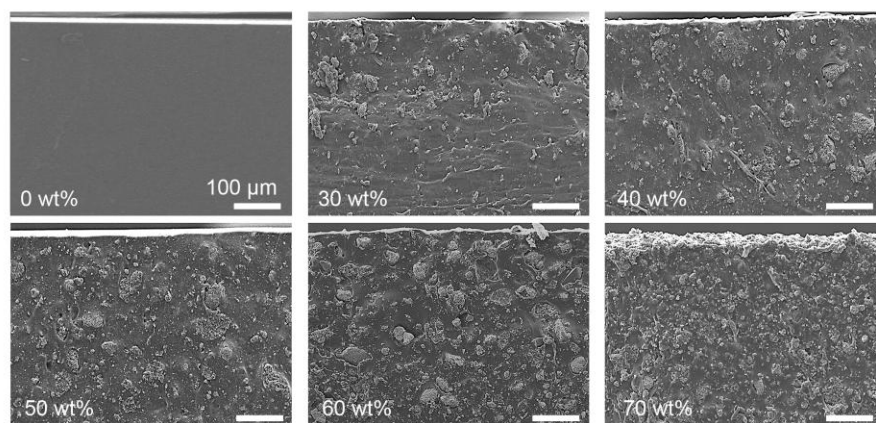


Fig. S5.

SEM images of PDMS and Cu-PDMS composite. Cross-sectional SEM images show Cu-PDMS composites with different weight ratios of Cu microparticles. The scale bars are 100 μm .

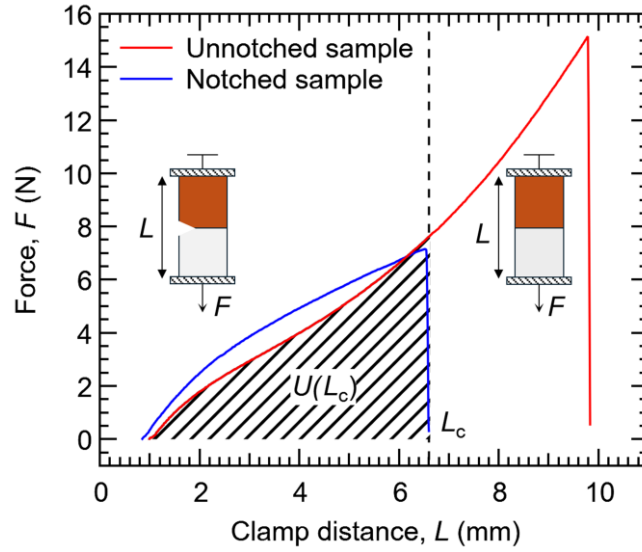


Fig. S6.

Experimental determination of the interfacial fracture energy. F – L curve curves of the unnotched (red) and notched (blue) to determine the fracture energy of Cu–PDMS/PDMS interface. An interfacial starter notch (~ 1 mm, sharp tip) was introduced along the Cu–PDMS/PDMS junction. The vertical dashed line indicates the critical clamp distance L_c taken from the notched test, where the interfacial notch begins to run. The hatched area under the red curve up to L_c represents the mechanical work delivered to the unnotched strip; normalizing this work by the initial width and thickness of the notched sample gives the interfacial fracture energy Γ . Insets illustrate the two test configurations and loading directions.

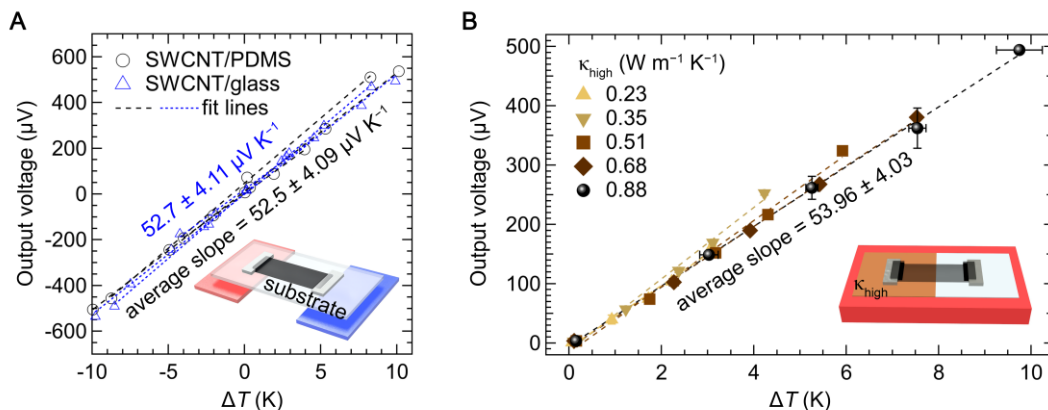


Fig. S7.

Output voltage of SWCNT as a function of temperature difference under different setups and substrates. (A) Output voltage of a p-type SWCNT film on a PDMS substrate measured using the conventional two-sided heating geometry (hot and cold Peltier modules imposing ΔT across the device). The extracted Seebeck slopes are $52.5 \pm 4.09 \mu\text{V K}^{-1}$ (SWCNT/PDMS) and $52.7 \pm 4.11 \mu\text{V K}^{-1}$ (SWCNT/glass). **(B)** Output voltage of SWCNT films supported on dual- κ substrate measured with a single heat source, where ΔT is induced laterally through κ contrast. Devices with different κ_{high} ($0.23\text{--}0.88 \text{ W m}^{-1} \text{ K}^{-1}$) show nearly identical slopes (average = $53.96 \pm 4.03 \mu\text{V K}^{-1}$), which is comparable to the intrinsic Seebeck coefficient of the SWCNT film within the experimental uncertainty. Error bars represent the standard deviation obtained from five independent devices.

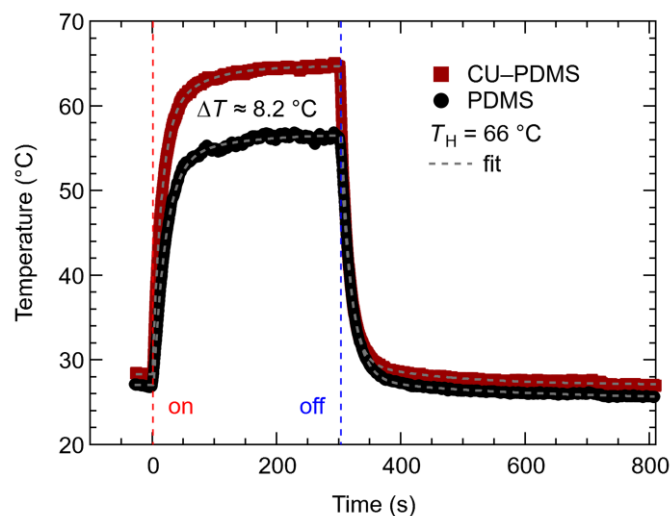


Fig. S8.

Surface temperature transients of Cu-PDMS and PDMS under heating-relaxation cycles.

The surface temperatures of Cu-PDMS and pristine PDMS were monitored under a constant heat source at $T_H = 66 \text{ }^{\circ}\text{C}$ using an infrared (IR) camera. The heating (on) and relaxation (off) switching times are indicated by the red and blue dashed lines, respectively. The experimental data (symbols) are well described by stepwise bi-exponential fits (grey dashed lines), capturing both the heating phase and the subsequent relaxation. The temperature difference stabilizes at $\Delta T_{\parallel} \approx 8.2 \pm 0.3 \text{ }^{\circ}\text{C}$, which is maintained robustly until t_{off} with no evidence of thermal drift.

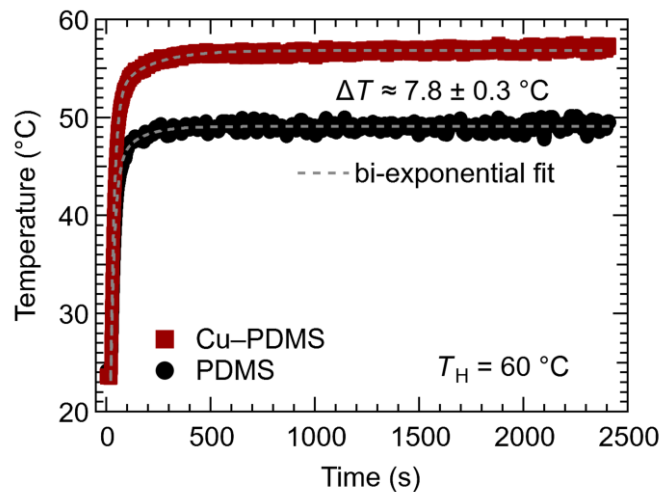


Fig. S9.

Long-term surface temperature stability of Cu-PDMS and PDMS. The surface temperatures of Cu-PDMS and pristine PDMS were monitored under a constant heat source at $T_H = 60 \text{ }^{\circ}\text{C}$ using an infrared (IR) camera. Both samples rapidly reach steady state, after which the temperature difference stabilizes at $\Delta T_{||} \approx 7.8 \pm 0.3 \text{ }^{\circ}\text{C}$. This stable temperature difference is maintained for more than 2000 s with no evidence of thermal drift.

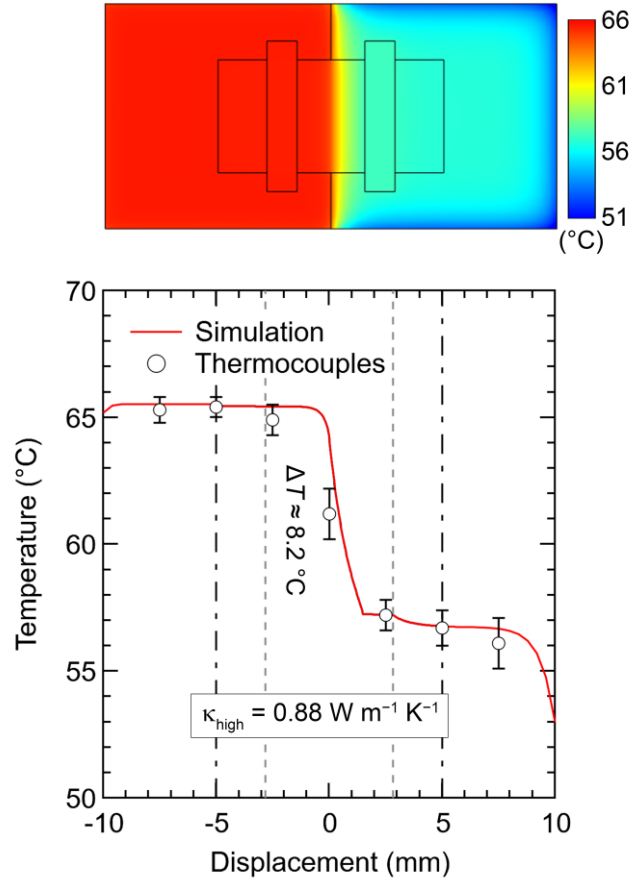


Fig. S10.

Thermal analysis of a single-cell pT-TEG device. (Top) FEA-simulated temperature distribution across the dual- κ substrate under bottom-side heating ($T_H = 66 \text{ } ^\circ\text{C}$). (Bottom) Comparison between in-plane temperature profiles obtained from FEA simulations and thermocouple measurements, demonstrating good agreement with an in-plane temperature drop of $\Delta T_{\parallel} \approx 8.2 \text{ } ^\circ\text{C}$ across the Cu–PDMS ($\kappa_{\text{high}} = 0.88 \text{ W m}^{-1} \text{ K}^{-1}$) and PDMS ($\kappa_{\text{low}} = 0.15 \text{ W m}^{-1} \text{ K}^{-1}$) interface. Data points represent the mean $\pm$ standard deviation from five independently fabricated devices.

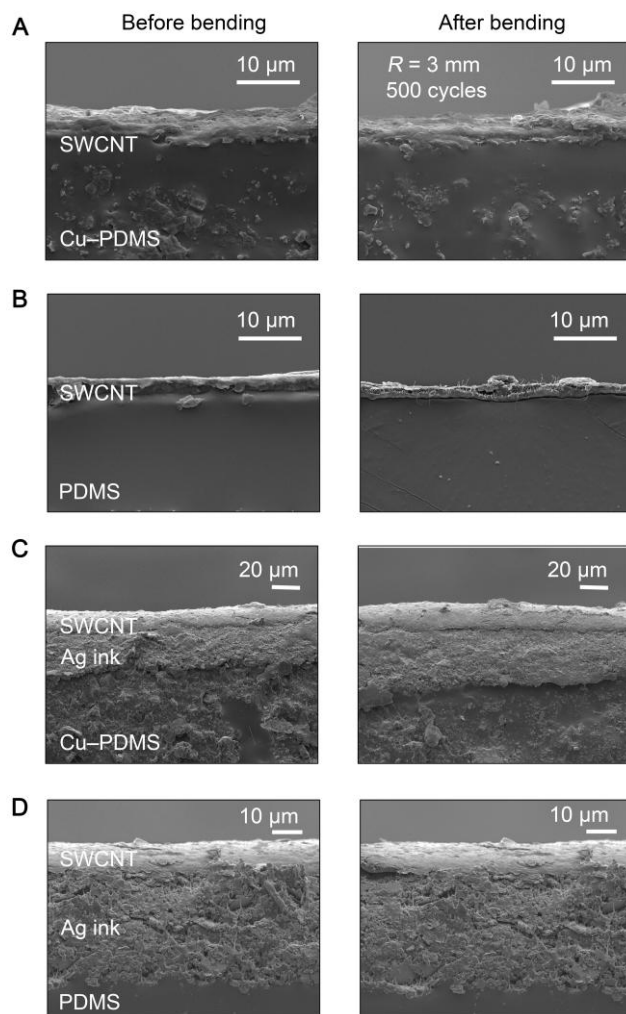


Fig. S11.

Cross-sectional SEM images of the substrate interfaces before and after bending. (A) SWCNT/ Cu-PDMS, (B) SWCNT/PDMS, (C) SWCNT/Ag ink/Cu-PDMS, and (D) SWCNT/Ag ink/PDMS structures before and after 500 bending cycles ($R = 3$ mm). All interfaces remained intact without visible cracks, delamination, or void formation after repeated bending, demonstrating the strong adhesion between the substrate, active and electrode layers. These results confirm that the electrical stability of the device is not affected by interfacial degradation, but is maintained by the mechanical integrity of the multilayer structure.

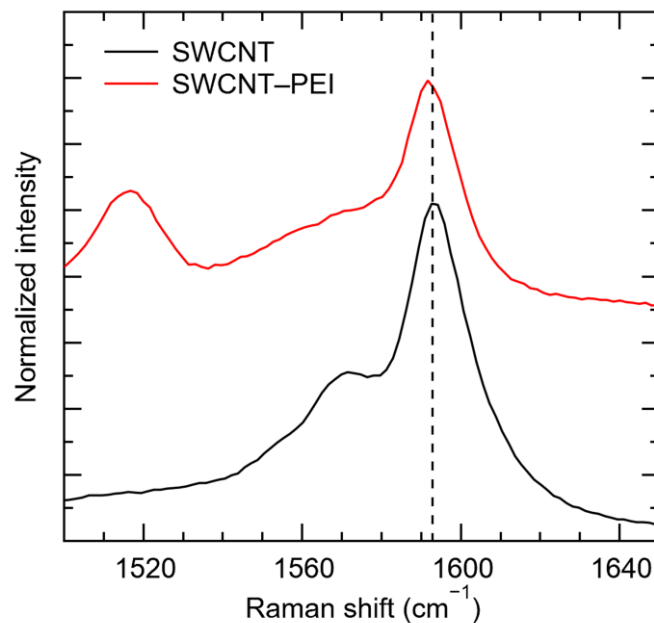


Fig. S12.

Normalized Raman spectra of SWCNT, and SWCNT-PEI films. The G-band peak for the SWCNT is observed at 1592.6 cm^{-1} . After treatment with polyethylenimine (PEI), the peak shifts to 1591.6 cm^{-1} . This downshift in the Raman frequency indicates charge transfer from PEI to SWCNT, suggesting n-doping of the SWCNT.

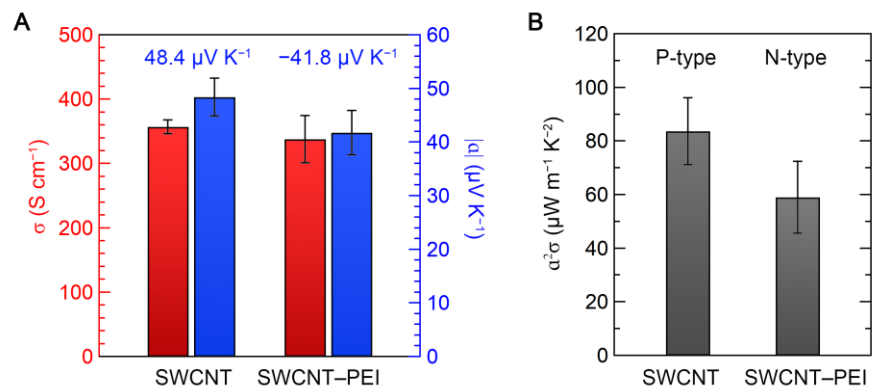


Fig. S13.

Thermoelectric properties of SWCNT and SWCNT-PEI films. (A) Electrical conductivity σ , Seebeck coefficient α and (B) power factor $\alpha^2\sigma$ for both SWCNT and SWCNT-PEI films.

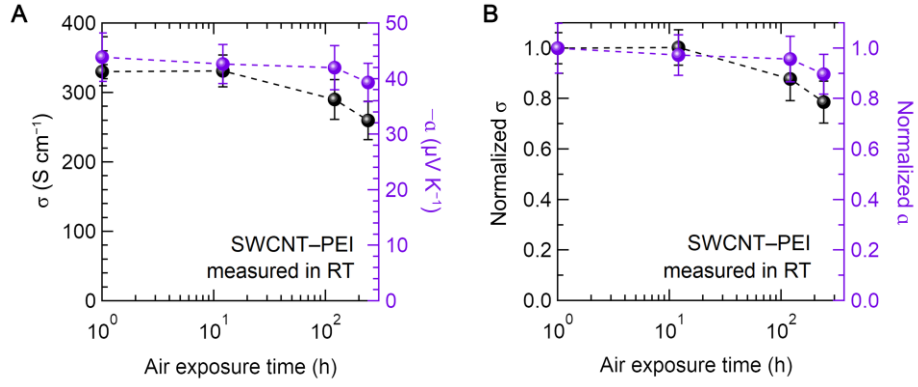


Fig. S14.

Ambient stability of PEI-doped SWCNT films. (A) Electrical conductivity σ , (black) and Seebeck coefficient α , (purple) as a function of air exposure time measured at room temperature. (B) Normalized σ and α , showing retention of ~100% after 12 h, ~88% after 120 h, and ~79% after 240 h of air exposure, indicating minimal degradation of TE properties over 100 h.

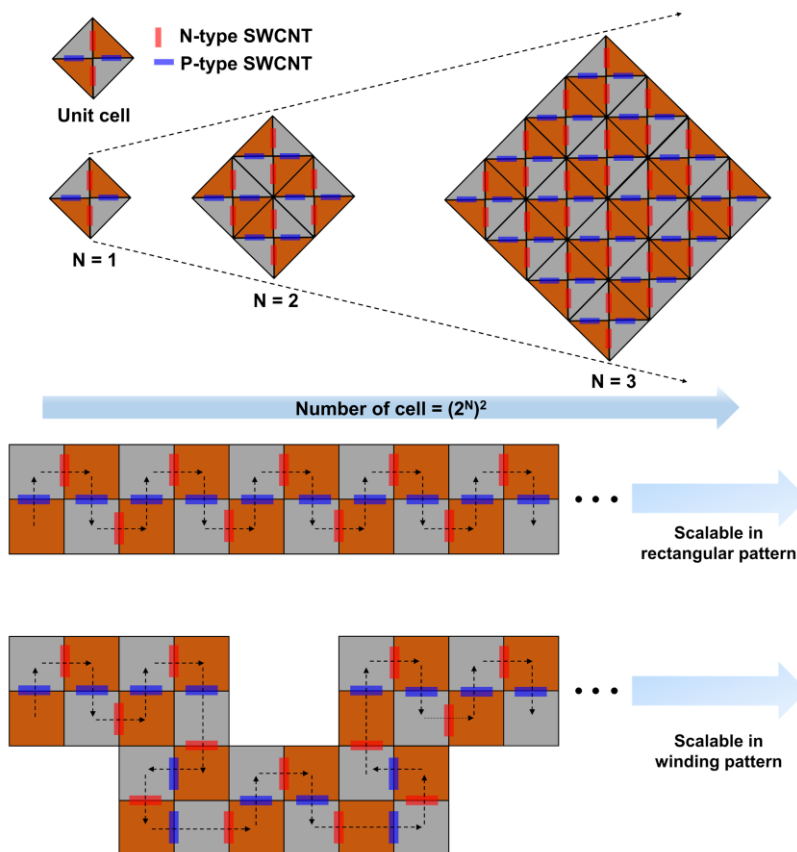


Fig. S15.

Scalable and versatile configurations of pT-TEG modules. The modules can be scaled by forming a chessboard-shaped square through repetition of N unit cells, each consisting of two p-type and two n-type SWCNT legs and assembling the chessboard-shaped units into arbitrary layouts, such as rectangular or meandering patterns, thereby demonstrating the structural adaptability of the design.

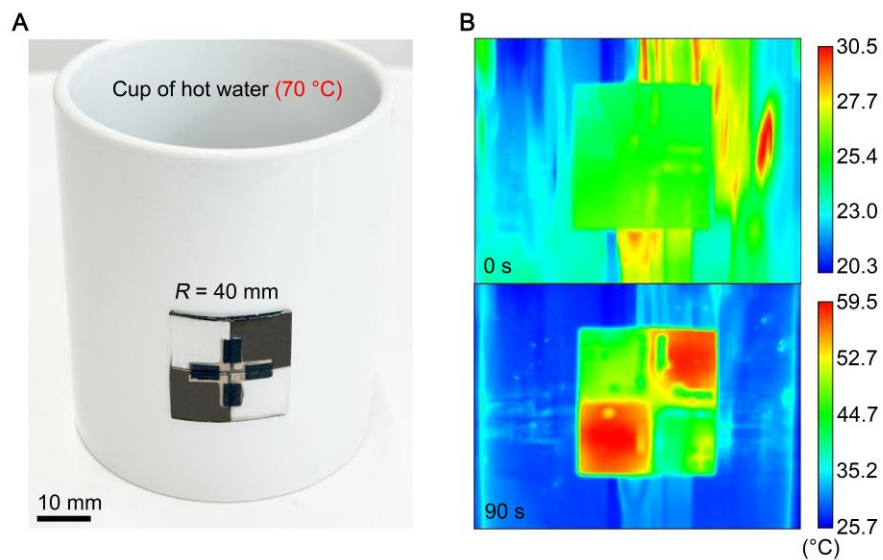


Fig. S16.

Stability of $\Delta T_{\parallel}$ in the bent state. (A) Photograph of a supplementary thermoelectric module attached to a cup containing hot water (70 °C, $R = 40 \text{ mm}$) under a bent condition. (B) Infrared thermal images of the module at 0 s and 90 s, demonstrating the stable temperature gradient during bending.

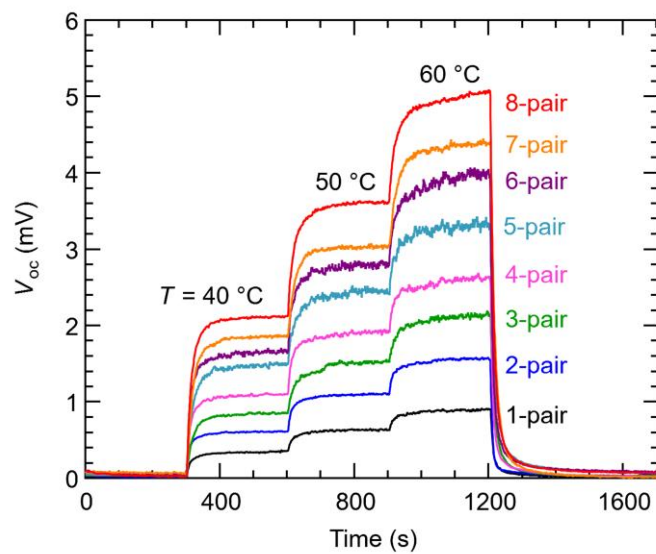


Fig. S17.

Open circuit voltage as a function of temperature and the number of modules. The V_{oc} values of the modules were measured over time, with heat source temperature ranging from 40 °C to 60 °C.

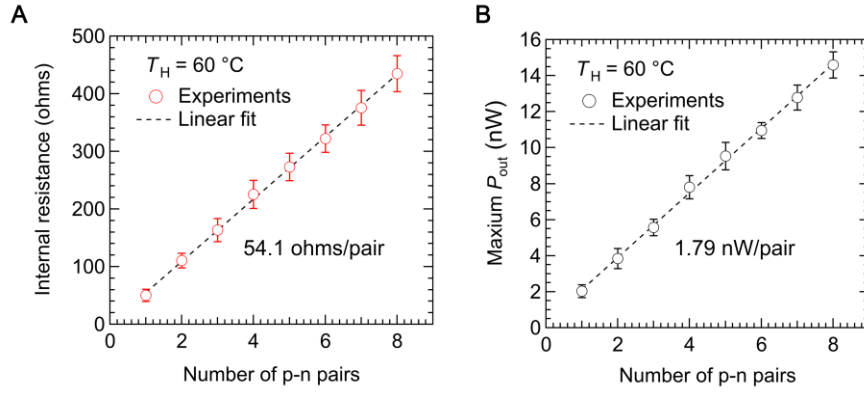


Fig. S18.

Internal resistance and maximum output power (P_{out}) of pT-TEG module as a function of the number of p–n pairs under $T_H = 60\text{ °C}$.

(A) Internal resistance increases linearly at approximately 54.1 ohms per pair, and (B) maximum P_{out} scales linearly at approximately 1.79 nW per pair, confirming ideal series behavior and negligible interfacial loss. Error bars represent standard deviations from three devices.

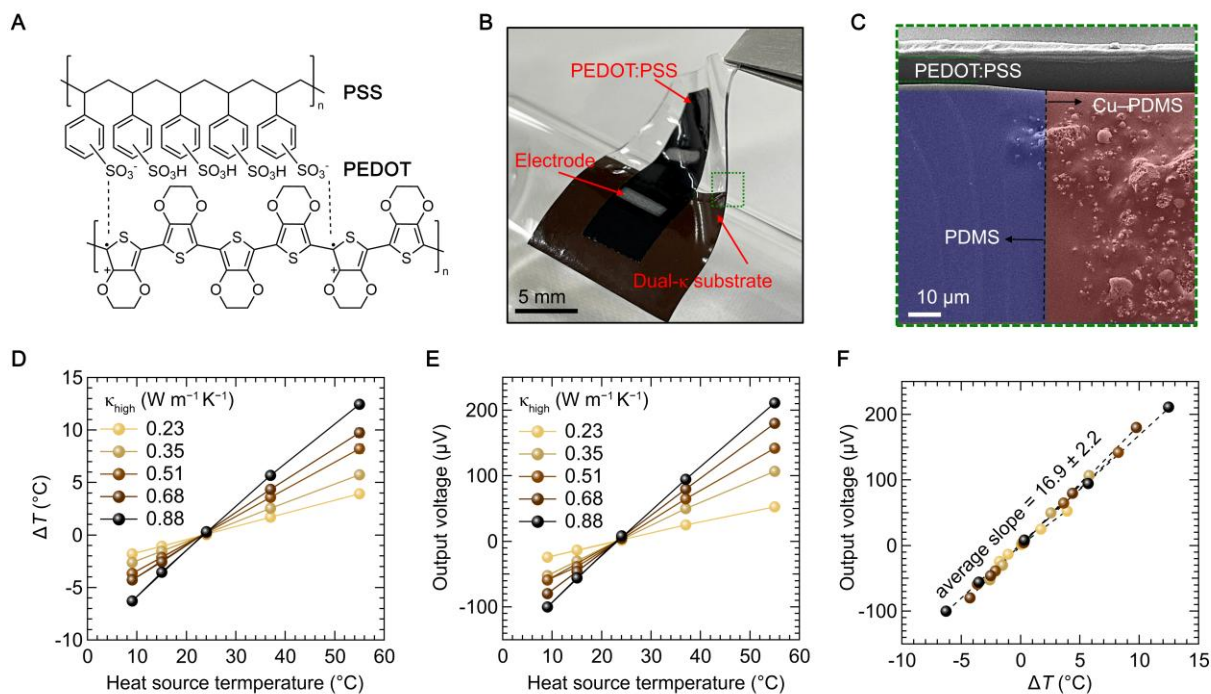


Fig. S19.

Demonstration of pT-TEG using commercially available PEDOT:PSS. (A) Chemical structure of PEDOT:PSS, (B) Photograph of the PEDOT:PSS-based pT-TEG device fabricated on a dual- κ substrate. (C) Cross-sectional SEM image showing the PEDOT:PSS layer on the dual- κ substrate (PDMS and Cu-PDMS). (D) Temperature difference across the PEDOT:PSS film as a function of the heat source temperature for substrates with different κ_{high} values. (E) Corresponding output voltage of the PEDOT:PSS pT-TEG device under the same conditions. (F) Linear relationship between ΔT and output voltage, yielding an average slope of $16.9 \pm 2.2 \mu\text{V K}^{-1}$, which is consistent with previously reported values for PEDOT:PSS (26).

Table S1.

Stepwise bi-exponential fitting parameters for surface temperature profiles of Cu–PDMS and PDMS. Parameters extracted from the stepwise bi-exponential fits shown in fig. S8. R^2 values indicate the goodness of fit for each regime.

Sample	T_{ss} (°C)	T_0 (°C)	a	τ_{1h} (s)	τ_{2h} (s)	τ_{1r} (s)	τ_{2r} (s)	R^2
PDMS	56.7	27.1	0.88	19.3	106.7	18.1	268.6	0.999
Cu–PDMS	64.7	28.3	0.81	13.7	62.8	13.6	145.3	0.999

Table S2.

Comparison of the pT-TEG with transverse thermoelectric materials. The table summarizes transverse thermopower, mechanical flexibility, processing temperature, and fabrication cost/scalability for this work and previously reported transverse thermoelectric platforms, including anomalous Nernst effect (ANE) and spin Seebeck effect (SSE) systems.

Materials	Mechanism	α ($\mu\text{V K}^{-1}$)	Flexibility	Processing temperature ($^{\circ}\text{C}$)	Cost / Process**	Reference
SWCNT (this work)	Seebeck (pT-TEGs)	$\sim 53^*$	Flexible	<100	Low (screen printing, solution)	This work
Mn ₃ Sn single crystal	ANE	0.35	Rigid	1080	High (Bridgman, low-throughput)	(66)
Co ₂ MnGa single crystal	ANE	6	Rigid	1250	High (arc/induction + slow cool)	(67)
Fe ₃ Pt/ MgO	ANE	2	Rigid	600–900	High (UHV MBE)	(68)
Co/Pt	ANE	1.0	Rigid	Room Temp.	Medium (HV sputter + e-beam)	(69)
Fe ₈₁ Al ₁₉	ANE	3.4	Flexible	Room Temp.	Medium (magnetron sputter, scalable)	(70)
Fe ₃ GeTe ₂	ANE	0	Rigid	-	High (CVT + exfoliation)	(40)
Fe ₃ Sn polycrystal bar	ANE	2.9	Rigid	805	High (arc melt + sealed anneal)	(41)
Fe ₃ Ga	ANE	4	Rigid	500	Medium (sputter + anneal)	(45)
Fe ₃ Al	ANE	2	Rigid	450	High (UHV MBE + anneal)	(45)
YMn ₆ Sn ₆ single crystal	ANE	2.0	Rigid	1000	High (Sn self-flux)	(48)
Ni ₈₁ Fe ₁₉ / Pt wire	SSE	0.022	Rigid	-	Medium (HV evaporation + sputter)	(44)

*the Seebeck response measured under a transverse geometry (not intrinsic ANE/SSE).

** Abbreviations in Process: UHV—ultrahigh vacuum. MBE—molecular beam epitaxy. HV—high vacuum. CVT—chemical vapor transport.

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